

WinSW Rclone Emulation Guide

Optional guide for using a generic WinSW + rclone mounted drive with emulator libraries and game directories. The mounted drive can come from any rclone-supported backend, including Google Drive, OneDrive, Dropbox, MEGA, S3-compatible storage, SFTP, WebDAV, and others.

1. Purpose

The WinSW Rclone service can mount a cloud or remote storage backend as a Windows drive letter. For emulation, this usually means placing large game files on the mounted drive while keeping performance-sensitive data local. The goal is convenience without turning the remote drive into a fragile replacement for every emulator folder.

2. Recommended Folder Model

Use the mounted drive for large read-mostly content. Keep saves, emulator configs, shader caches, firmware, and active install data on a local SSD whenever possible.

Folder type	Recommended location	Reason
Game/ROM archives	Mounted drive, for example R:\Games	Large, mostly read-only files are a good fit.
Extracted disc images	Mounted drive or local SSD	Use local SSD for games that stream heavily.
Saves	Local SSD	Avoid sync delay and write-cache risk.
Emulator configs	Local SSD	Small files, should remain available offline.
Shader caches	Local SSD	Frequent writes and latency-sensitive reads.
Firmware/keys	Local SSD	Private and emulator-specific.
DLC/updates	Local SSD or mounted drive	Depends on emulator and update format.

3. Example Directory Layout

```
R:\Games\  
  Switch\  
  WiiU\  
  PS2\  
  PS3\  
  Xbox360\  
  
C:\EmulationData\  
  Saves\  
  Configs\  
  ShaderCache\  
  Firmware\  
  Keys\
```

The exact emulator names do not matter. The important part is separating large read-only content from small write-heavy or private data.

4. Cache Guidance

The default service template uses full VFS cache mode. This improves compatibility with programs that expect normal disk behavior, but it consumes local disk space.

Use case	Suggested cache size
Small retro libraries	40G to 80G
Modern console images	120G to 250G
Very large game libraries	250G or more if local space allows

A game larger than the cache limit can still work if it does not need the entire game cached at once. For better stability, set the cache size at least as large as the biggest game or image you use often, plus extra room.

5. Suggested Service Options For Game Libraries

The default template already includes settings that favor large-file reads and long directory caching. A typical profile includes:

```
--vfs-cache-mode full
--vfs-cache-max-size 120G
--vfs-cache-max-age 7d
--buffer-size 512M
--vfs-read-ahead 2G
--dir-cache-time 720h
--poll-interval 30s
--network-mode
```

Increase `--vfs-cache-max-size` if your local disk has enough free space and your games are larger than the current cache limit.

6. What To Keep Local

- Save files and save states.
- Shader caches and pipeline caches.
- Emulator configuration files.
- Firmware, keys, account files, or other private emulator data.
- Games currently being installed, patched, compressed, or converted.

7. What Can Live On The Mounted Drive

- ROMs, ISOs, WUX/WUD, CHD, RVZ, XCI, NSP, and other large game files, where legally owned.
- Compressed backups and archival copies.
- Box art, metadata, and static media if your frontend tolerates network latency.
- Completed converted files that are no longer being actively written.

8. Frontend And Emulator Tips

- Point the emulator or frontend game directory to the mounted drive, for example `R:\Games\Switch`.
- Keep the emulator itself installed locally.
- Do not put the emulator portable profile on the mounted drive unless you accept slower startup and possible sync delays.

- For launchers, scan only the platform folders you actually use to avoid long indexing times.
- If a game stutters during heavy streaming, test a local copy to confirm whether the remote mount is the bottleneck.

9. Safety Notes

- Do not store secrets, keys, or private account files in a public repository or shared drive.
- Do not commit rclone.conf, logs, OAuth tokens, or generated service XML with personal paths.
- Remote-mounted game libraries depend on internet speed, provider limits, and local cache health.
- Use local backups for save data. Do not rely on VFS cache as a backup system.

10. Verification Checklist

- The Windows service is running.
- The drive letter is visible in File Explorer.
- The cache folder has enough free space.
- The emulator sees the game directory.
- A small game launches before testing a very large title.
- Saves are written to a local folder and backed up separately.

This guide is optional. The WinSW Rclone installer remains a general-purpose Windows service template for rclone mounts.